News Topic Classification

Group 2

Team Members:

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Motivation

1. Automated news classification is essential for search engines and recommendation systems
2. Most studies evaluate on a single dataset — how sensitive are results to dataset characteristics?
3. How does dataset structure affect the effectiveness of contextual embeddings vs. bag-of-words features?

Core Question

When does contextual language modeling (BERT) actually provide benefits over traditional keyword-based methods?

Agenda

1 **Data Pipeline & Dataset Preparation**
Ming-Hsiang Lee

2 **Baseline Model — TF-IDF + Logistic Regression**
Ruoxuan Chen

3 **BERT Fine-Tuning**
Xinyan Huang

4 **Evaluation & Comparative Analysis**
Meiling Wang

5 **Conclusion & Discussion**
Tzu-Chieh Yang

Data Preparation & Data Pipeline

Data Download · Preprocessing · Dataset Migration

Data Preparation

01	**Data Download**	Wrote python script to fetch datasets from HuggingFace.
02	**Data Preprocessing**	Built full preprocessing pipeline: text cleaning, data splits, column schema, debug subset
03	**Dataset Migration**	Switched from BBC News to AG News. Rewrote both scripts, designed split strategy
04	**Team Infrastructure**	Committed all data to repo, and documented in README

Timeline & Milestones

Week	Proposed Plan	Actual Progress
Week 1	Finalize dataset and complete preprocessing	**ON TRACK** **BBC News pipeline complete** All processed CSVs in repo so team can use immediately
Week 2	Develop baseline classifier using n-gram features and Logistic Regression	**AHEAD** **Baseline + BERT both done** Both pipelines committed in 1 week Proposal planned these with 2 weeks
Week 3	Fine-tune BERT/DistilBERT and compare results against the baseline	**ON TRACK** **Evaluation & error analysis added** BBC test accuracy ~97% (baseline) Results solid — but 1.2k articles felt too small for BERT
Week 4	Conduct error analysis, consolidate findings, and prepare the final presentation	**EXPANDED** **Expanded scope to AG News** Migrated pipeline to AG News (127k articles) Baseline + BERT both updated

Datasets

	BBC News	AG News
Role	Initial prototype	**Full benchmark**
Size	~1.2k articles	~127k articles
Categories	5	4
Label set	Business, Sport, Tech, Entertainment, Politics,,	Business, Sports, Sci/Tech, World
Train	835 rows (70%)	102,000 rows (85%)
Val	179 rows (15%)	18,000 rows (15%)
Test	180 rows (15%)	7,600 rows **(official)**

Data Preprocessing

Raw Data

1
load_dataset()

Auto-detect CSV
or folder layout

2
clean_text()

Strip non-ASCII
Keep original casing

3
preprocess()

Drop nulls/dupes
Add **label_id** & **text_lower**

4
split_dataset()

BBC News: 70/15/15
AG News: 85/15+ official test

5
make_debug_subset()

10 samples/class
Smoke-testing subset

6
save_splits()

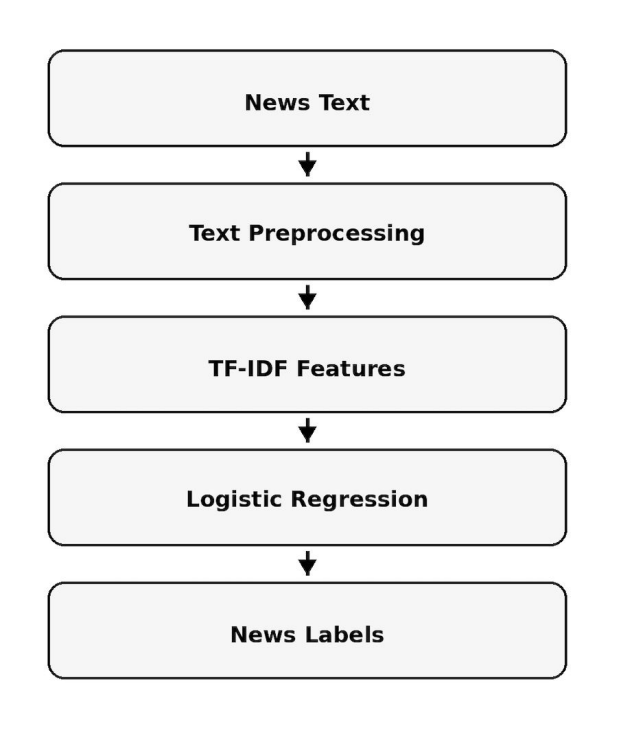
Write train/val/test
& debug CSVs

Processed CSVs

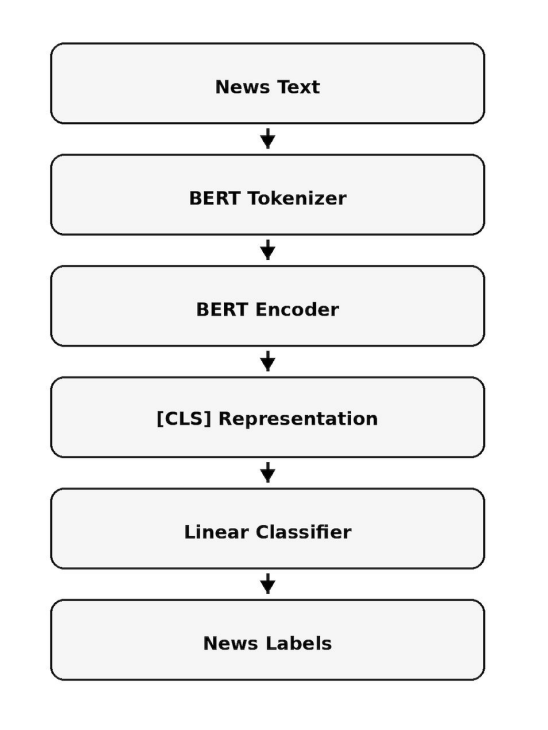
Baseline Model（TF-IDF + LR）
&
BERT Fine-Tuning

Method Comparison

Baseline



Bert

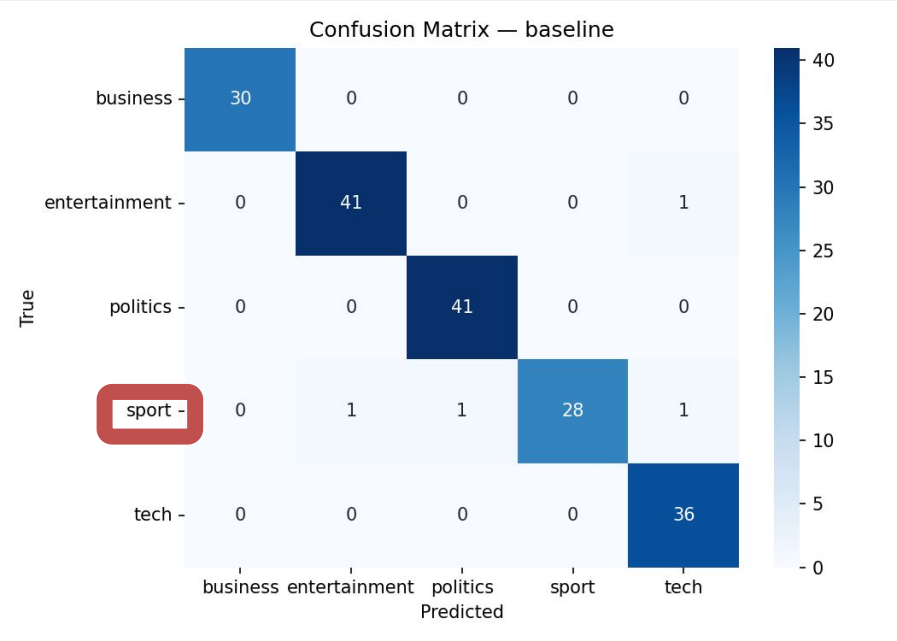


Evaluation & Comparative Analysis

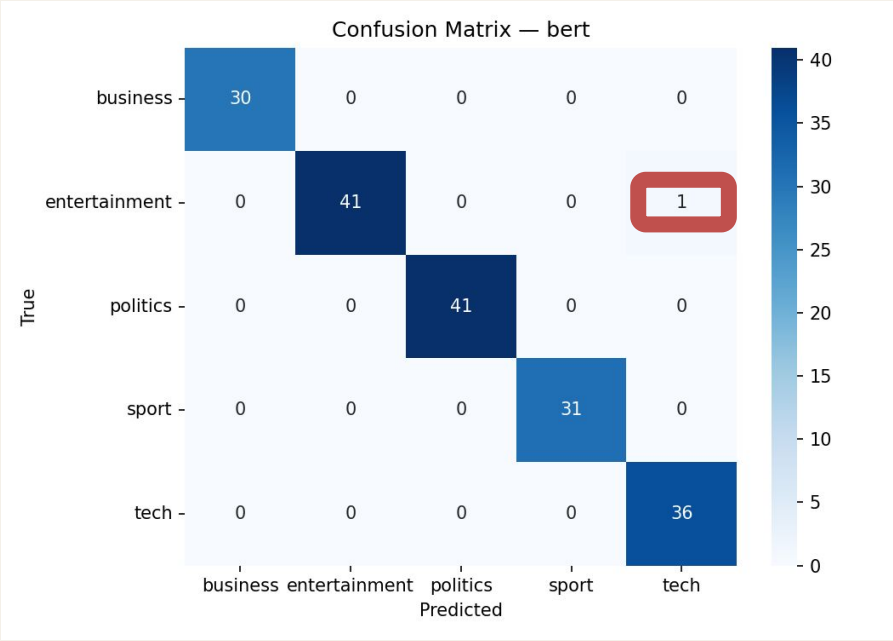
Baseline (TF-IDF + LR) vs BERT Fine-Tuning

BBC News: Baseline and Bert

Baseline



Bert



Error Analysis – BBC News

BBC News Baseline error + BERT both error

True label: Entertainment | Predicted: Tech
"Crossrail link to get go-ahead... UK Treasury has allocated £7.5bn for the project... fast cross-London rail link... link Heathrow Airport to Canary Wharf... reduce journey time to 40 minutes... support from government and business groups..." – An infrastructure/transport article about a £10bn rail project.

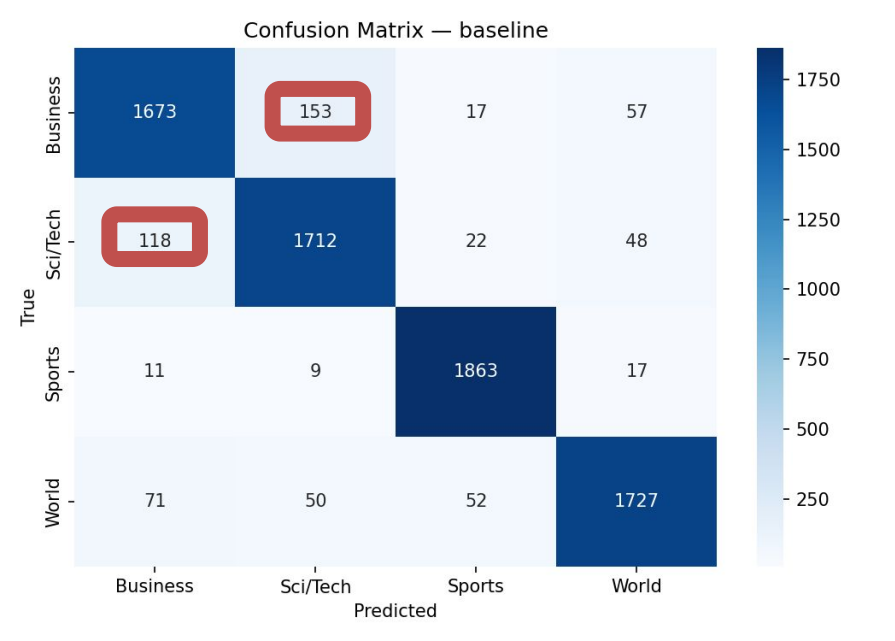
Reason: Category Ambiguity + Dataset Bias

- Contains infrastructure and transport-related content
- Keywords like "*rail link*", "*project*", "*transport*" → may be associated with **Tech**
- However, the dataset labels it as Entertainment, which is not clearly aligned with the content

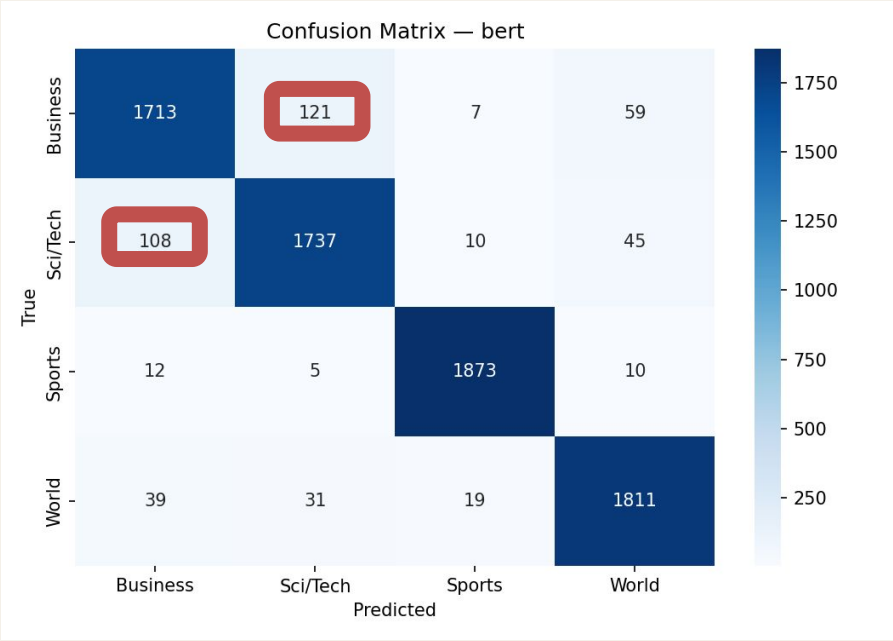
Insight:
This example shows that unclear or inconsistent category definitions can lead to systematic misclassification.

AG News: Baseline and Bert

Baseline

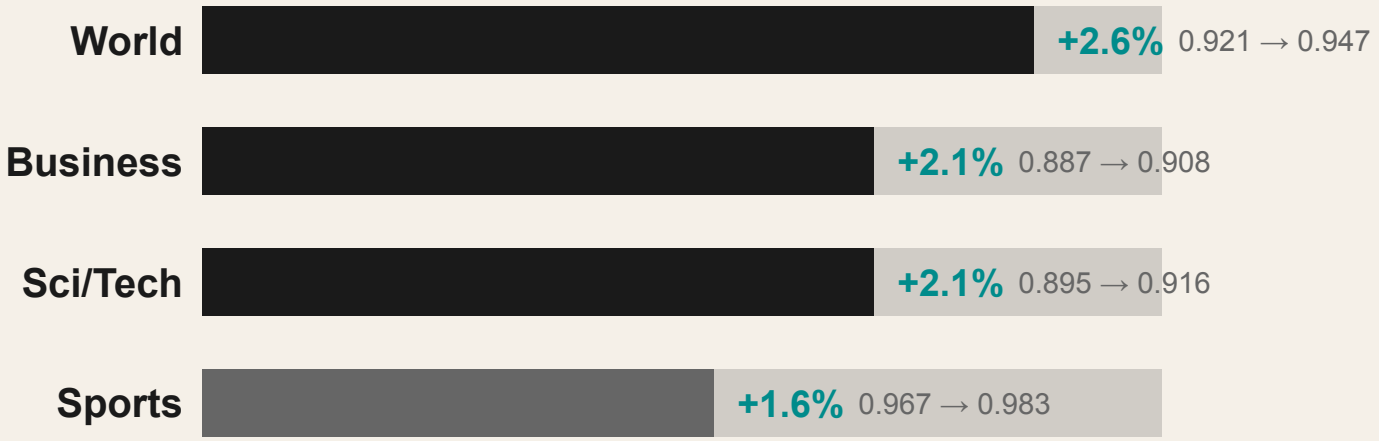


Bert



Where Does BERT Help Most?

Per-class F1 Gain — AG News



Insight: BERT gains most on ambiguous classes (World, Business, Sci/Tech) where context matters. Sports barely improves — TF-IDF already captures its distinctive vocabulary.

Error Analysis – AG News

AG News Baseline + BERT both error

True label: Sci/Tech | Predicted: Business
“IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.”

AG News Baseline + BERT both error

True label: Business | Predicted: Sci/Tech
“Broadband charges set to tumble The cost of broadband internet access is likely to fall after Ofcom ordered British Telecom to cut the amount it charges internet providers.”

Reason: Category Overlap

- Contains both technology company context and business-related signals
- Tech aspect: computing giant, IBM
- Business aspect: hiring, headcount growth

Insight:
The boundary between Sci/Tech and Business is ambiguous.

Results at a Glance

Focus on the delta — what the gap tells us, not the raw numbers.

	Baseline	BERT	Δ Gain
BBC News — Accuracy	97.78%	99.44%	**+1.66%**
BBC News — Macro F1	0.977	0.994	**+0.017**
AG News — Accuracy	91.78%	93.87%	**+2.09%**
AG News — Macro F1	0.917	0.939	**+0.022**

Key Finding: BERT's gain is larger on the harder dataset (AG News +2.09%) than the easier one (BBC +1.66%). BERT's advantage scales with task difficulty.

BBC vs AG News: Why the Gap?

Same models, different datasets — the gap reveals dataset effects.

AG News

120,000 samples × 4 classes

Short headlines (~8 words avg)
Overlapping topics (Biz ↔ Sci/Tech)
Harder boundary → more room for BERT

Best Accuracy: 93.87%

BBC News

2,225 samples × 5 classes

Full articles (~380 words avg)
Well-separated categories
Easier → baseline already near-perfect

Best Accuracy: 99.44%

Takeaway: AG News requires understanding context beyond keywords, while BBC News can often be solved using distinctive vocabulary — this is why BERT shows more benefit on AG News

Discussion

01 Errors are systematic, not random

Business ↔ Sci/Tech: 153 errors (baseline) → 121 (BERT) — the #1 confused pair in both models.

02 Dataset matters more than model choice

Same BERT: 93.87% (AG News) vs 99.44% (BBC) — 5.6% gap from data, not architecture.

03 TF-IDF and BERT fail differently

TF-IDF fails when categories share keywords. BERT fails less, but still on short ambiguous headlines.

Conclusion

Two pipelines — TF-IDF + LR vs BERT — tested on AG News & BBC News

01

Data > Model

Dataset complexity drives the performance gap more than model architecture.

02

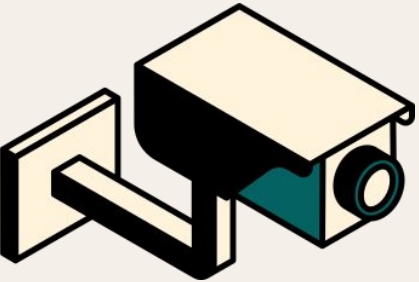
BERT Helps When It's Hard

Biggest gains on ambiguous classes where keyword matching falls short. Limited gains when vocabulary is already distinctive.

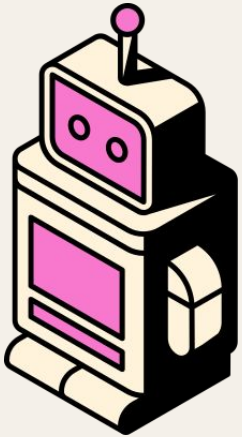
03

Errors Point to Label Quality

Most misclassifications come from overlapping categories, not model weakness.



Thank You



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